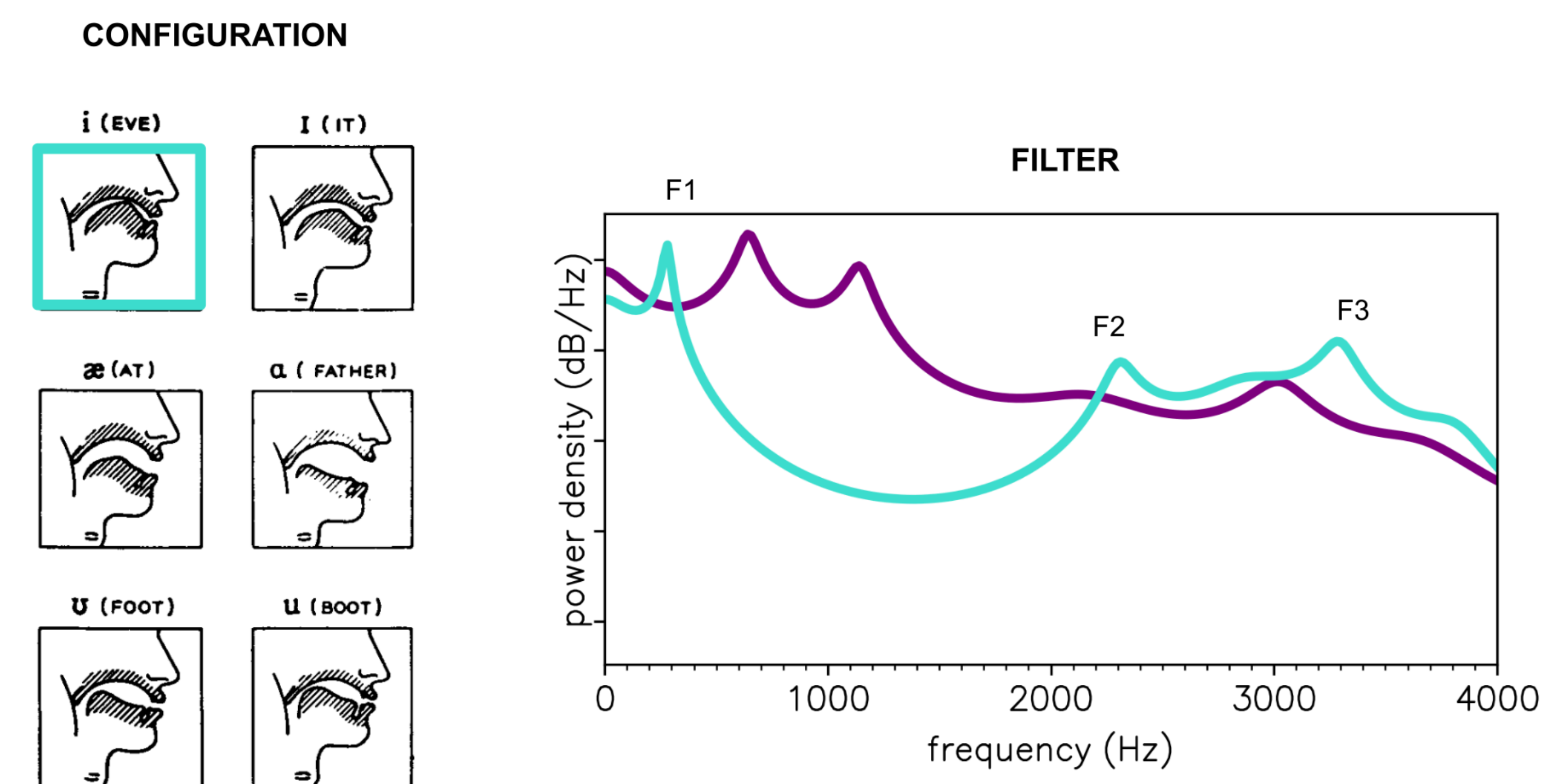


TELL YOUR FILTER WHAT YOU ALREADY KNOW

A FILTER IS A POLYNOMIAL

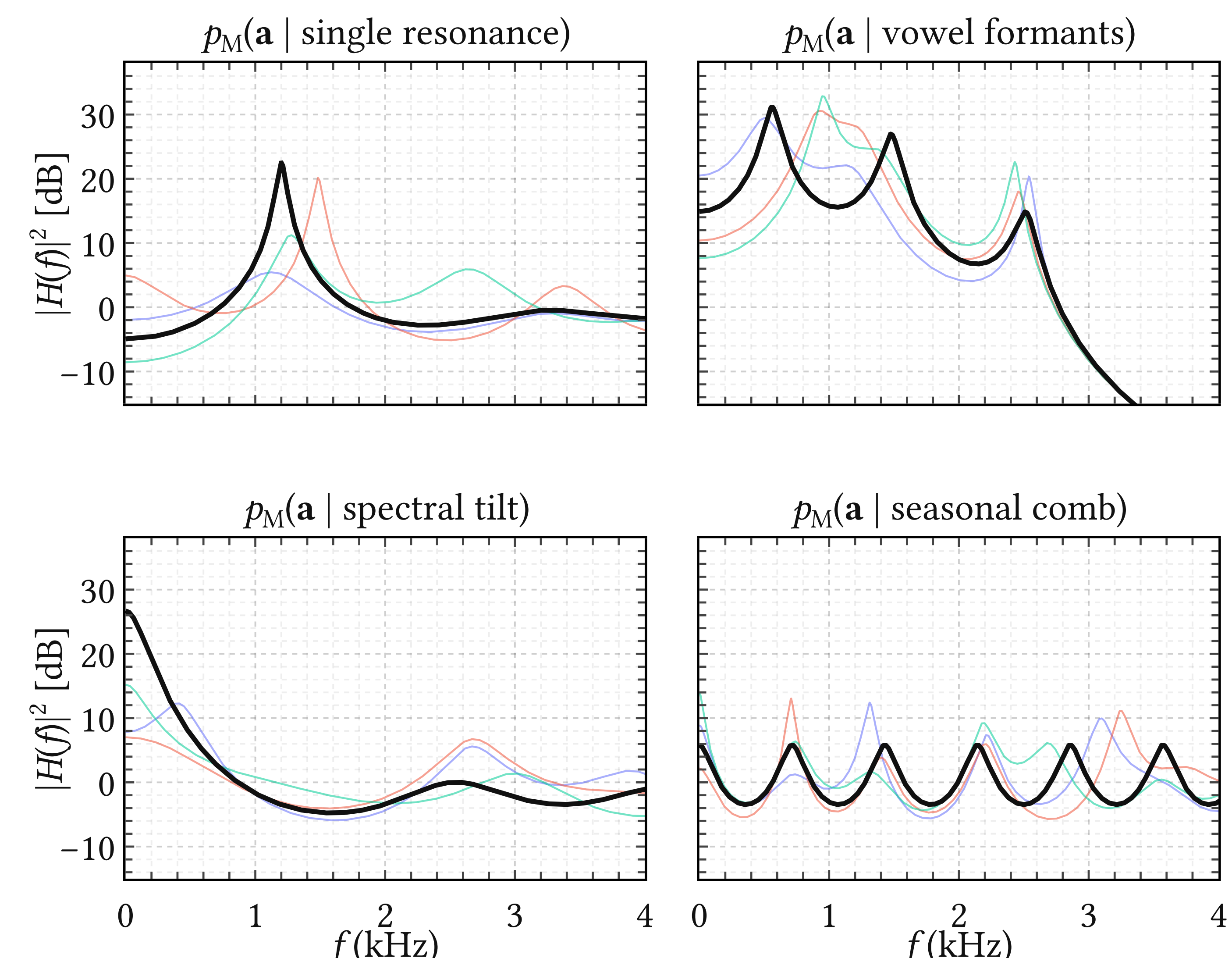
The vocal tract is an all-pole filter: P numbers, fitted from 20 ms of speech. Every resonance you expect is a factor of that polynomial.



Two vowels, two filters (Rabiner & Schafer).

ONE UPDATE, FOUR BELIEFS

Prior over the filter before (grey) and after (black) telling it about a resonance, three formants, a spectral tilt, a harmonic comb. Same closed-form update every time.

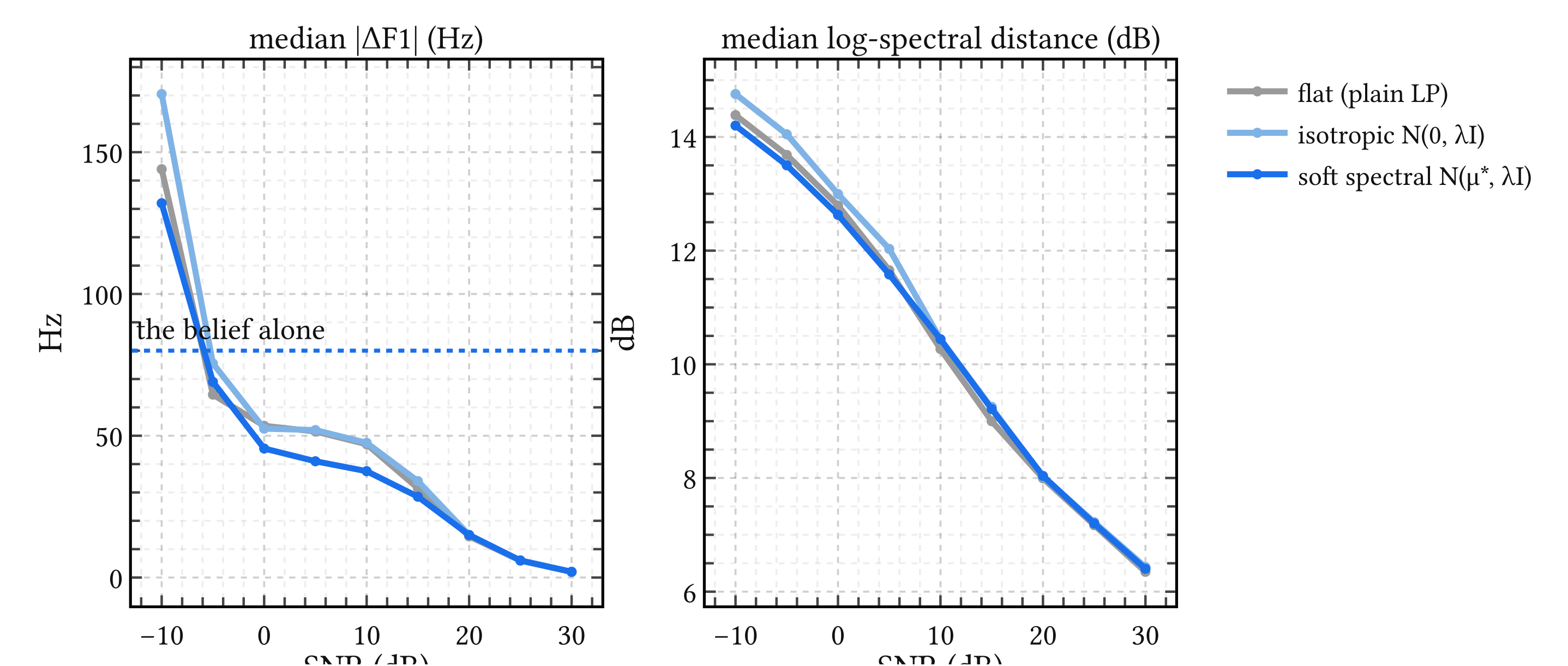
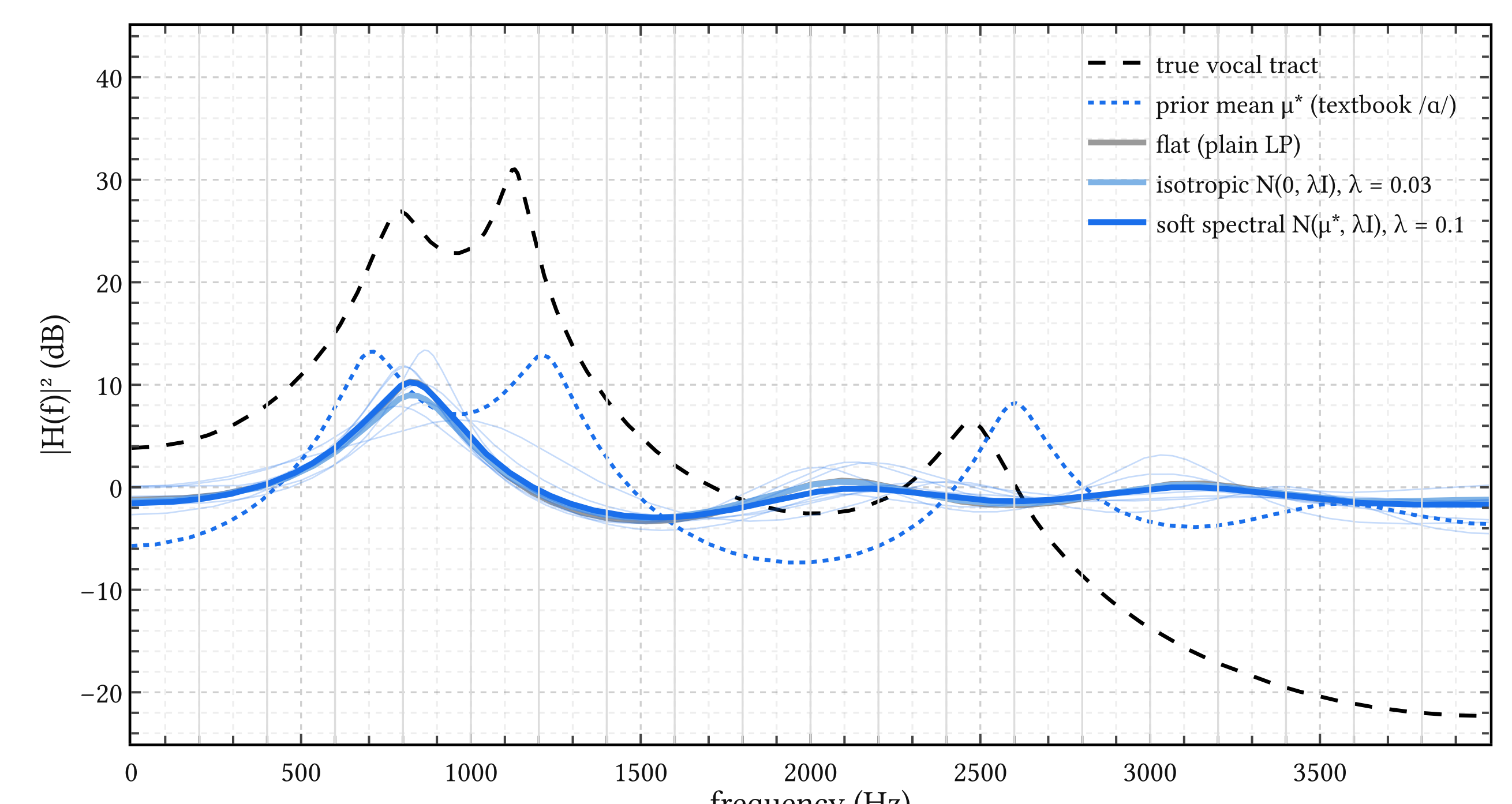


LEAST CHANGE OF BELIEF

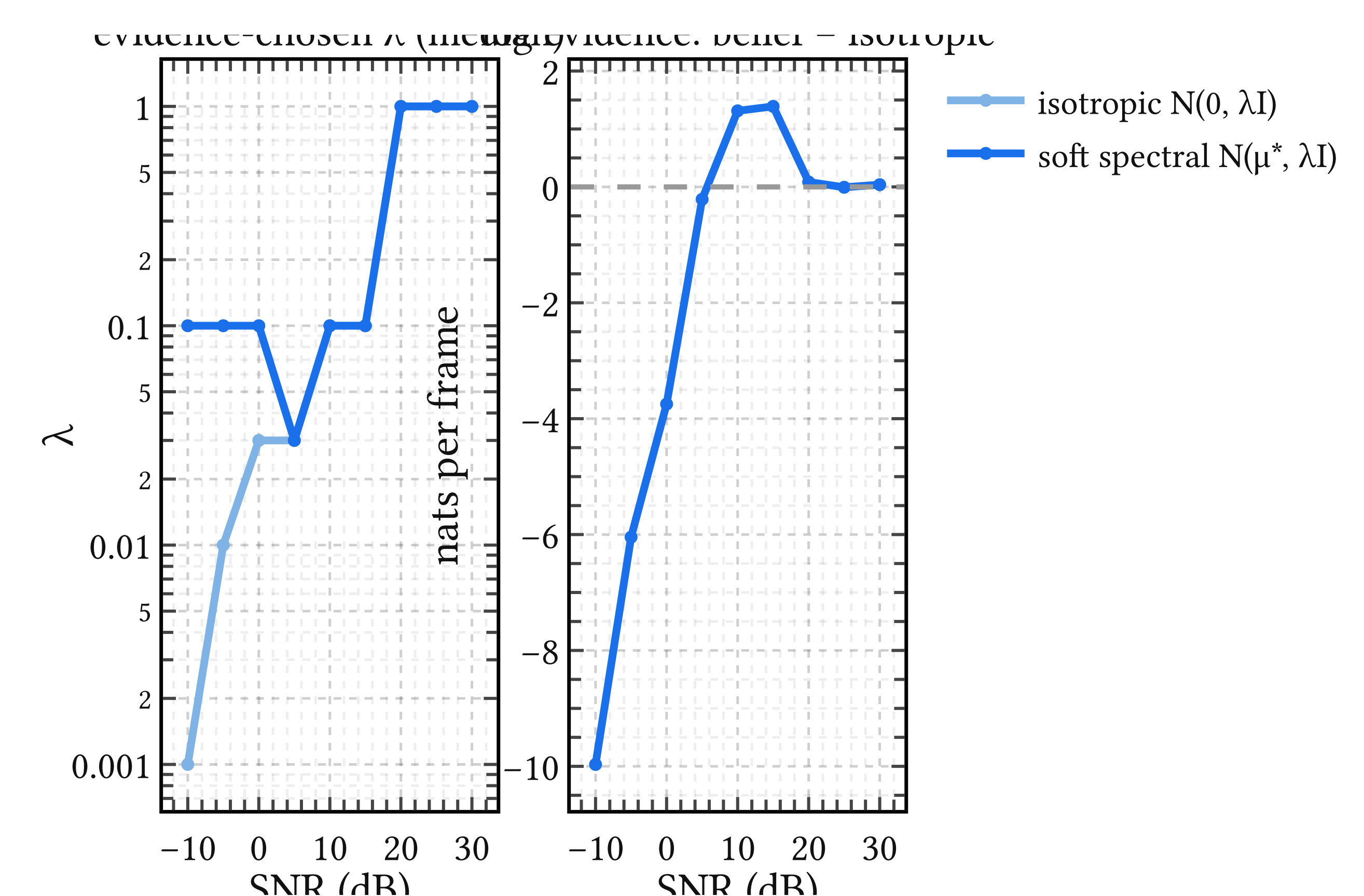
Move the prior as little as possible (D_{KL}) so that the polynomial is divisible by the feature *on average*. Only the mean moves, and only where the belief lives.

IN NOISE, IT HELPS

Synthetic /a/, 20 ms frames. The belief is the *textbook* /a/, 10% off this speaker. How much to trust it is chosen per frame by the evidence.



Below 15 dB the belief cuts the first-formant error by a fifth; above 25 dB nothing is lost.



The data decide: the evidence trusts the belief in noise and switches it off in clean frames.

